```
   return
centroids
      #6
```

#1 Method `randomCentroidInit` takes the list of points in the dataset and the number of centroids it should create. It returns the list of centroids generated.
#2 Initialize the list of centroids by randomly sampling (without replacement) `numCentroids` points from the dataset.
#3 Cycle through the list of centroids (their indices).
#4 For each centroid, cycle through its coordinates (assuming `dim` , the number of coordinates, is f.i. a class variable, otherwise you could use `|centroids[i]|`).
#5 Update the current coordinate by adding some random noise.
#6 Return the list of centroids.

As mentioned, we abstracted away the logic of the update and score functions in separate methods. But before looking at those, it's interesting to check out the random initialization step, that's often underestimated. Listing 12.2 shows the implementation we suggest for this function: while there are several viable alternatives (for instance, randomly drawing each coordinate from the domain's boundaries, or using actual dataset's points), a solution that gets us several advantages is randomly perturbating `k` points casually drawn from the dataset:

· First, we don't have to worry about the domain's boundaries: if each centroid's coordinate was generated completely at random, we would have to